

# Fangyi Chen

Senior Research Scientist @ Intelligent Creation, ByteDance

<https://fangyi-chen.github.io/>

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## Education

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- Ph.D. in Electrical and Computer Engineering 2020-2025  
Carnegie Mellon University Pittsburgh, USA
- M.S. in Electrical Engineering 2017-2018  
University of Pittsburgh Pittsburgh, USA
- B.E. in Electrical Engineering and Its Automation 2013-2017  
North China Electric Power University Beijing, China

## Research Summary

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- Perception & Multimodal Foundation Models: Core contributor to ByteDance’s Vidi 3.0 foundation model for long-term video understanding and the TikTok video editing agent (2026). Extensive publication record in object detection, segmentation, image editing, few-shot, weakly-supervised, open-vocabulary learning, and CNN/Transformer architectures (2020-2026).
- Large-Scale Data Engines: Established the world’s largest dataset for retail scene product detection, recognition, OCR, and deployed robots across 350 Walmart stores (2021). Designed robust data engines to power open-vocabulary object detection (2023), advanced image editing via RefLade 1.0 (2024), reasoning segmentation (2025), and video scene graph generation (2026).

## Publications ([Google Scholar](#))

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- [1] Fangyi Chen, Yaojie Shen, Lu Xu, Ye Yuan, Shu Zhang, Yulei Niu, Longyin Wen. Referring Layer Decomposition. *International Conference on Learning Representations (ICLR)*, 2026.
- [2] Chen Li, Zhantao Yang, Han Zhang, Fangyi Chen, Chenchen Zhu, Anudeepsekhar Bolimera, Marios Savvides. MetaVLA: Unified Meta Co-training For Efficient Embodied Adaption. *International Conference on Learning Representations (ICLR)*, 2026.
- [3] Chen Li, Han Zhang, Zhantao Yang, Fangyi Chen, Zihan Wang, Anudeepsekhar Bolimera, Marios Savvides. STELAR-VISION: Self-Topology-Aware Efficient Learning for Aligned Reasoning in Vision. *AAAI Conference on Artificial Intelligence (AAAI)*, 2026.
- [4] Hao Chen, Yujin Han, Fangyi Chen, Xiang Li, Yidong Wang, Jindong Wang, Ze Wang, Zicheng Liu, Difan Zou, Bhiksha Raj. Masked autoencoders are effective tokenizers for diffusion models. *International Conference on Machine Learning (ICML)*, 2025.
- [5] Hao Chen, Ze Wang, Xiang Li, Ximeng Sun, Fangyi Chen, Jiang Liu, Jindong Wang, Bhiksha Raj, Zicheng Liu, Emad Barsoum. Softvq-vae: Efficient 1-dimensional continuous tokenizer. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [6] Fangyi Chen, Han Zhang, Zhantao Yang, Hao Chen, Kai Hu, Marios Savvides. RTGen: Generating Region-Text Pairs for Open-Vocabulary Object Detection. 2024.
- [7] Fangyi Chen, Han Zhang, Kai Hu, Yu-kai Huang, Chenchen Zhu, Marios Savvides. Enhanced Training of Query-Based Object Detection via Selective Query Recollection. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

- [8] Fangyi Chen, Han Zhang, Zaiwang Li, Jiachen Dou, Shentong Mo, Hao Chen, Yongxin Zhang, Uzair Ahmed, Chenchen Zhu, Marios Savvides. Unitail: Detecting, Reading, and Matching in Retail Scene. *European Conference on Computer Vision (ECCV)*, 2022.
- [9] Chenchen Zhu, Fangyi Chen, Uzair Ahmed, Zhiqiang Shen, Marios Savvides. Semantic Relation Reasoning for Shot-Stable Few-Shot Object Detection. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.
- [10] Chenchen Zhu, Fangyi Chen, Zhiqiang Shen, Marios Savvides. Soft Anchor-Point Object Detection. *European Conference on Computer Vision (ECCV)*, 2020.
- [11] Fangyi Chen, Chenchen Zhu, Zhiqiang Shen, Han Zhang, Marios Savvides. NCMS: Towards Accurate Anchor Free Object Detection through l2 Norm Calibration and Multi-Feature Selection. *Computer Vision and Image Understanding (CVIU)*, 2020 Jul 27:103050.
- [12] Han Zhang, Fangyi Chen, Zhiqiang Shen, Qiqi Hao, Chenchen Zhu, Marios Savvides. Solving Missing-Annotation Object Detection With Background Recalibration Loss. *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2020.
- [13] Fangyi Chen, Chenchen Zhu, Marios Savvides. A Novel Collaborate Control Strategy for Enhanced Training of Vehicle Recognition. *The IEEE 90th Vehicular Technology Conference (VTC)*, 2019.
- [14] Ker-Jiun Wang, Kaiwen You, Fangyi Chen, Prakash Thakur, Michael Urlich, Soumya Vhasure, and Zhi-Hong Mao. Development of Seamless Telepresence Robot Control Methods to Interact with The Environment Using Physiological Signals. *The 13th ACM International Conference on Human-Robot Interaction (HRI)*, 2018.
- [15] Ker-Jiun Wang, Anna Zhang, Kaiwen You, Fangyi Chen, Quanbo Liu, Yu Liu, Zaiwang Li, Hsiao-Wei Tung, and Zhi-Hong Mao. Ergonomic and Human Centered Design of Wearable Gaming Controller Using Eye Movements and Facial Expressions. *The IEEE International Conference on Consumer Electronics (ICCE)*, 2018.
- [16] Ker-Jiun Wang, Kaiwen You, Fangyi Chen, Zihang Huang, and Zhi-Hong Mao. Human-Machine Interface Using Eye Saccade and Facial Expression Physiological Signals to Improve the Maneuverability of Wearable Robots. *The International Symposium on Wearable & Rehabilitation Robotics (WeRob 2017)*, 2017.

## Patents

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|---------------------|-----------------------------------------------------------------------------------------|
| [1] US12189714B2    | System and method for improved few-shot object detection using dynamic semantic network |
| [2] US20250182450A1 | System and method for weapon detection with pose estimation                             |
| [3] US12266156B2    | System and method for solving missing annotation object detection                       |
| [4] US20250005881A1 | System and method for assigning complex concave polygons as bounding boxes              |
| [5] US12131497B2    | Fast object search based on the cocktail party effect                                   |
| [6] US20240355085A1 | System and method for matching products and determining spreads and plugs               |
| [7] US11915463B2    | System and method for the automatic enrollment of object images into a gallery          |
| [8] WO2020210825A1  | System and method for detecting products and product labels                             |
| [9] WO2022211995A1  | System and method for using non-axis aligned boxes for retail detection                 |
| [10] WO2022169622A1 | Soft anchor point object detection                                                      |
| [11] WO2022109295A1 | System and method for detecting and classifying abnormal cells                          |
| [12] US11954175     | Feature pyramids for object detection                                                   |
| [13] US2022058432A1 | Few-shot object detection using semantic relation reasoning                             |

## Professional Experience

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Senior Research Scientist in Generative AI @ Intelligent Creation, ByteDance 07/2025 - present

- Video Understanding: Drove the post-training of ByteDance's Vidi video understanding model.
- Image Understanding and Editing: Spearheaded the research of Referring Layer Decomposition, enabling MLLMs to perform open-world image manipulation.
- Video Agent: Drove the post-training of the TikTok Editing Agent utilized for video editing.
- TikTok Live Effects: Primary developer of TikTok's real-time portrait matting algorithm.

Research Assistant & Associate III @ CyLab, CMU 02/2019-05/2025

- Research: Scene understanding, training strategy, and vision transformer
- U.S. Department of Defence, Project Maven: Training strategy with insufficient and imperfect data.
- Walmart Robot: System and robot for store management and automatic checkout in 350 stores.

Research Intern @ ByteDance, Bellevue 05/2024-08/2024

- Multimodal LLM: Conducted research on multi-modal large language models for in-depth image editing, decomposition, and synthesis.

Journal Reviewer:

- IEEE Transactions on Image Processing (since 2022)
- ELSEVIER Pattern Recognition (since 2020)
- IEEE Transactions on Geoscience and Remote Sensing (since 2020)
- ELSEVIER NeuralComputing (since 2022)
- Springer Visual Computer (since 2022)
- Connected Science (since 2022)
- Multimedia Systems (since 2020)
- Springer International Journal of Computer Vision (since 2025)

Conference Reviewer:

- The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2023, 2024, 2025
- The Annual Conference on Neural Information Processing Systems 2024, 2025
- International Conference on Computer Vision 2023, 2025
- European Conference on Computer Vision 2024
- The International Conference on Learning Representations 2023, 2024, 2025
- International Conference on Machine Learning 2023, 2024, 2025

## Awards

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- Carnegie Institute of Technology Dean's Fellowship, Carnegie Mellon University 2020
- Best Hardware Hack, PITT-CTSI GitHub Major League Hacking 2018
- National 3rd Prize, Chinese National College Students Competition on Energy Economics 2015
- University Merit Student, NCEPU 2014